

CORRECTIVE AND PREVENTIVE ACTION REPORT

CAPA-2028-014: Violet detector QC failures at high room temperature

CAPA number	CAPA-2028-014
Status	Closed, effective
Record type	CAPA report (ISO 13485 8.5.2, 8.5.3; 21 CFR 820)
Opened / closed	2028-07-14 / 2028-12-15
Priority	High (product performance, HA-08)
CAPA owner	Atif Lodhi, Systems Engineering
Product	CSI-100 with 405 nm laser option; software 3.0.1

Summary

Field reports of daily QC failures on the violet (405 nm) detector array at room temperatures above 27 °C. Root cause: supplier firmware change in the 405 nm laser module not notified, combined with fan control that did not follow laser temperature. Corrections, corrective and preventive actions verified effective.

CAPA Record Summary

Field	Entry
CAPA number / title	CAPA-2028-014: Violet detector QC failures at high room temperature
Product	CSI-100 Cell Sorting Instrument, configurations with the 405 nm laser (CFG-CSI-001 3-, 4-, 5- and 6-laser builds); software 3.0.1
Source	Complaints CMP-2028-0031, -0038, -0044, -0052, -0057, -0061, -0066; service reports; post-market trend of instrument internal logs (RMF-CSI-001 Section 16)
Opened	2028-07-14 by Quality Engineering (trend signal exceeded threshold)
Priority / risk	High: product performance; related hazard HA-08 (incorrect result from QC or calibration)
CAPA owner	Systems Engineering (A. Lodhi)
Team	Systems Eng, Optics R&D, Firmware, Supplier Quality, Service, Clinical Affairs, Regulatory, Quality
Related records	ANM-0062 (ADL-CSI-001); ECO-0241, -0244, -0246; CR-SW-0512, -0514, -0521; SCAR-2028-009; QA-L1 Rev C; SRS-CSI-001 Rev E; SRS-SW-CSI-010 Rev F; RMF-CSI-001 Rev D
Closed	2028-12-15 after effectiveness check met all criteria

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1. Problem Description

1.1 Problem statement

Between 2028-05-02 and 2028-07-10, 7 customer sites reported that the daily performance check failed, or passed with warning, on the violet (405 nm) detector array of the CSI-100. Failures occurred mostly in the afternoon and on days when the laboratory was warm. Because the QC lockout (RA-803) blocked IVD assays after a failed check, laboratories lost instrument time and delayed testing until the QC passed on a repeat run or a service visit. No incorrect patient result has been reported.

1.2 Is / Is not

	Is	Is not
What	Violet array (405 nm) QC failure or warning: robust CV up 30 to 60%, intensity down 8 to 15%	Other lasers (488, 561, 633 nm) or scatter parameters
Where	7 of 64 installed instruments with the 405 nm laser; all 7 with laser serial numbers from lot L1-2801 or later	Instruments with 405 nm lasers from earlier lots (L1-2704 to L1-2712)
When	Room temperature 27 to 30 °C; typically 3 to 6 h after power-on	Below 26 °C; first QC of the morning
How many	41 QC failures or warnings in 9 weeks across 7 sites	Complaints of incorrect results: none
Since	Instruments shipped from 2028-02 onward	Instruments shipped 2027-11 to 2028-01

1.3 Signal detection

The post-market trend of instrument internal logs (FW-701 to FW-704), reviewed quarterly, showed the 405 nm laser base temperature at up to 36 °C on affected units, against 31 °C on units from earlier lots at the same room temperature. Combined with the complaint count, the signal exceeded the CAPA trigger in the post-market surveillance procedure (3 complaints of the same failure mode in 90 days).

2. Risk Assessment, Containment and Reportability

2.1 Health hazard evaluation

The failure affects detection sensitivity on the violet array, which is related to HA-08 (incorrect result from configuration, QC or calibration) in RMF-CSI-001. The existing risk controls worked as intended: the QC lockout prevented IVD assays from running after a failed check, and assay controls were in range on every run released. A performance drift that develops between a passing QC check and a later run could, however, reduce sensitivity for dim violet-excited markers during the day. Clinical Affairs reviewed released results from the 7 sites: no result differed from a repeat run beyond precision limits. Estimated risk: S4, P2 (review region), unchanged from the residual risk in RMF-CSI-001, with lost instrument availability as the main effect.

2.2 Containment and corrections

Date	Action	Owner	Status
2028-07-15	Hold of 14 laser modules from lots L1-2801 to L1-2806 in inventory and in service stock	Supply Chain	Done
2028-07-18	Customer technical bulletin TB-CSI-2028-02 to all sites with 405 nm lasers: keep room below 26 °C where possible, repeat QC after a failure, contact service if it fails twice	Service, Regulatory	Done
2028-07-22	Temporary service procedure: extra fan duty setting on affected units	Service	Done (superseded by CR-SW-0512)
2028-07-25	Clinical review of released results from affected sites (Section 2.1)	Clinical Affairs	Done: no impact

2.3 Reportability

Regulation	Decision	Rationale
21 CFR 803 (MDR)	Not reportable	No death or serious injury; the malfunction would not be likely to cause or contribute to death or serious injury if it recurred, because the QC lockout and assay controls detect it before results are reported
21 CFR 806 (corrections and removals)	Not reportable; record kept	Firmware update and bulletin improve performance and availability; they are not made to reduce a risk to health, which the existing controls already address
EU IVDR vigilance (Art. 82, 83)	Not a serious incident; included in trend reporting	Same rationale; trend documented in the periodic safety update

Decisions reviewed and signed by Regulatory Affairs on 2028-07-26 and re-confirmed at closure.

3. Investigation

3.1 Data collected

Data	Finding
Complaint and service records (7 sites)	All failures on 405 nm; all lasers from lots L1-2801 or later; room 27 to 30 °C
Internal logs from 64 instruments	Laser base temperature 34 to 36 °C on new-lot units vs 30 to 31 °C on old-lot units at 28 °C room; fan speed identical
Returned laser modules (3)	Output power and pointing stable at 25 °C base; at 35 °C base, power fell 12% and beam pointing moved 6 µm in the vertical axis
Supplier data (Vendor L1)	Lot L1-2801 introduced new TEC controller firmware that lowers cooling current to extend TEC life; not listed in any change notification
Incoming inspection records	Power and beam quality tested at 22 °C only; firmware version not recorded
Design verification (DVP-CSI-301, -403)	Operating temperature test at 30 °C passed on DV units, all with lasers from lot L1-2611 (old firmware)
ANM-0062 (validation, 2027)	QC warning on violet array at Site 1 on a 29 °C day; closed as bead lot and room temperature; laser base temperature was not examined

4. Root Cause Analysis: Fishbone, 5-Why and Fault Tree

4.1 Fishbone (Ishikawa) diagram

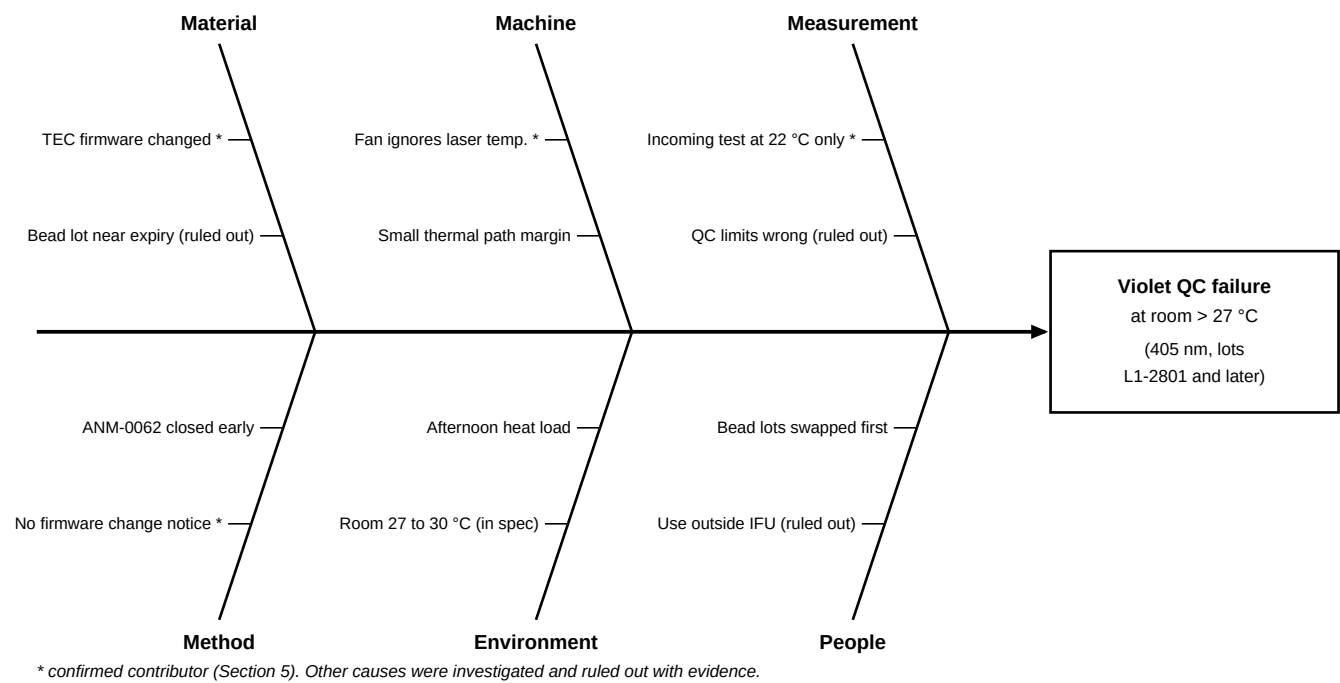


Figure 1. Fishbone diagram. Causes marked * were confirmed by test; the others were ruled out.

4.2 5-Why analysis

Two chains were analyzed: the technical cause of the failure, and the systemic cause of why it reached the field.

	Chain A: technical	Chain B: systemic
Why 1	Why did the violet QC fail? Violet signal CV rose and intensity fell during the day.	Why did affected lasers reach customers? Incoming inspection and design verification did not detect the change.
Why 2	Why did the signal degrade? 405 nm laser power fell up to 12% and the beam spot moved 6 μ m.	Why did inspection not detect it? Incoming test is at 22 °C and does not record firmware version.
Why 3	Why did power and pointing change? Laser base temperature reached 35 to 36 °C, above its 32 °C stabilization limit.	Why was the firmware change unknown? Vendor L1 did not notify it.
Why 4	Why did base temperature rise? New TEC firmware (lot L1-2801) reduced cooling current; fan speed stayed fixed.	Why was there no notification? Quality agreement QA-L1 Rev B required notice of hardware and process changes only, not firmware.
Why 5	Why did the fan not compensate? Fan control follows room temperature, not laser temperature, and was designed with margin for the original laser behavior.	Why did the agreement exclude firmware? The supplier agreement template predates embedded firmware in purchased modules; supplier controls were not reviewed against it.
Root cause	Insufficient thermal margin: fan control not tied to laser base temperature, exposed by a supplier firmware change.	Supplier change control did not cover firmware in purchased components.

A third, contributing finding came from the review of ANM-0062: the 2027 validation investigation was closed on a plausible explanation without data that ruled out the instrument. The anomaly procedure did not require environmental causes to be reproduced or excluded before closure.

4.3 Fault tree

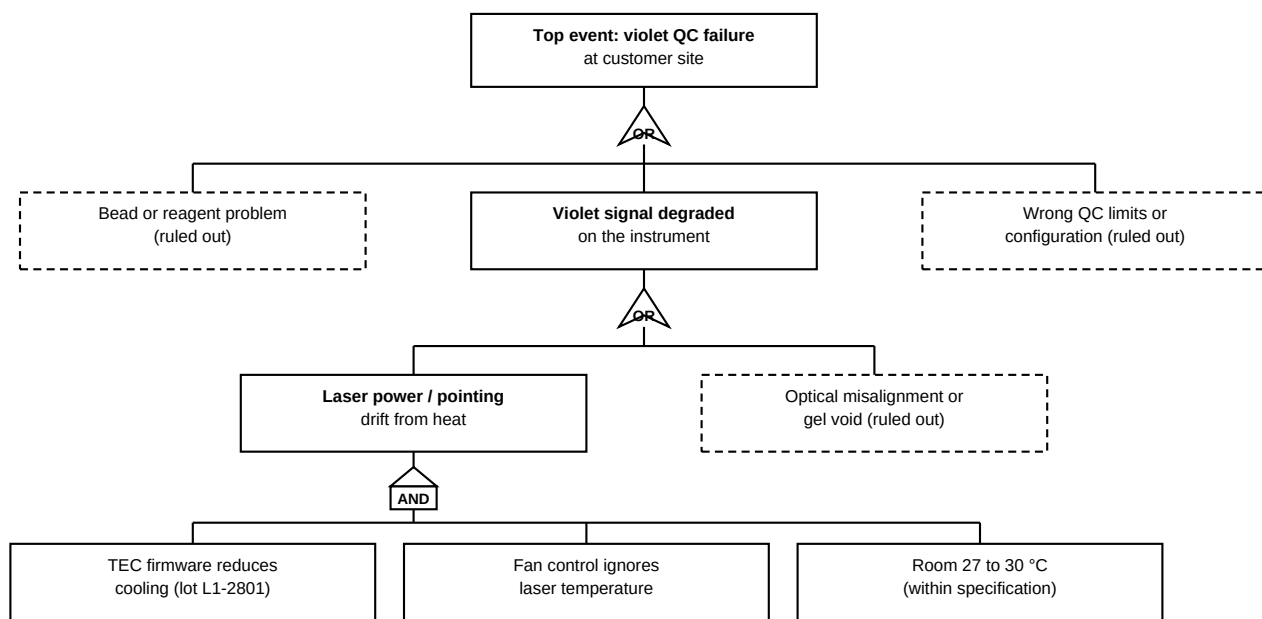


Figure 2. Fault tree. Dashed branches were investigated and ruled out. All three basic events under the AND gate are needed for the failure; removing any one prevents it.

5. Root Cause Confirmation

The root cause was confirmed by reproducing the failure and by showing that removing each contributing cause prevents it (protocol CAPA-2028-014-T1, environmental chamber, instrument DV-03 refurbished to production standard).

Test	Laser lot	Room	Fan control	Laser base temp.	Violet QC (6 h)	Result
1 Baseline	L1-2611 (old firmware)	30 °C	Room-based (3.0.1)	31.2 °C	Pass, rCV +4%	No failure
2 Reproduce	L1-2803 (new firmware)	30 °C	Room-based (3.0.1)	35.8 °C	Fail at 3.5 h, rCV +52%	Failure reproduced
3 Room below limit	L1-2803	25 °C	Room-based (3.0.1)	30.4 °C	Pass, rCV +3%	No failure
4 Firmware restored	L1-2803 reflashed to old TEC firmware	30 °C	Room-based (3.0.1)	31.5 °C	Pass, rCV +5%	No failure
5 New fan control	L1-2803	30 °C	Laser-temperature based (3.0.2 candidate)	29.8 °C	Pass, rCV +2%	No failure
6 Worst case	L1-2803	32 °C (above spec)	Laser-temperature based	31.6 °C	Pass, rCV +6%	Margin confirmed

Tests 2 to 5 confirm that the failure needs the new firmware, the room-based fan control and a warm room together, matching the AND gate in Figure 2. Test 6 shows the corrected design keeps the laser below its 32 °C limit even 2 °C above the specified operating range.

6. Root Cause Statement

Code	Level	Root cause
RC-D	Direct cause	The 405 nm laser base temperature exceeded its 32 °C stabilization limit at room temperatures of 27 to 30 °C, causing power loss and beam pointing drift.
RC-1	Design root cause	Fan control followed room temperature instead of laser base temperature, leaving no thermal margin when the laser's cooling behavior changed; design verification used lasers from a single lot.
RC-2	Systemic root cause 1	Supplier change control did not cover firmware in purchased modules: quality agreement QA-L1 Rev B and the supplier agreement template excluded firmware changes; incoming inspection did not check firmware version or test at elevated temperature.
RC-3	Systemic root cause 2	The anomaly procedure allowed ANM-0062 to be closed without reproducing or excluding environmental causes, and service troubleshooting pointed to bead lots first, so an early signal was missed.

Actions in Section 7 are coded to the root cause they address. 'Extent' marks preventive actions that apply the same lesson to other components, suppliers, processes or products.

7. Action Plan

The plan has 27 actions: 4 corrections (fix the problem found), 11 corrective actions (remove the root causes) and 12 preventive actions (prevent the same kind of problem elsewhere). Design changes went through design control; production changes through engineering change and design transfer.

7.1 Action summary

ID	Type	Action	Root cause	Owner	Due	Completed
C1	Correction	Field firmware update 3.0.2 to installed base	RC-D, RC-1	Service	2028-09-30	2028-09-26
C2	Correction	Disposition of held laser lots	RC-D	Supply Chain	2028-09-15	2028-09-12
C3	Correction	New QC baselines after firmware update	RC-D	Service	2028-10-07	2028-09-30
C4	Correction	Retrospective review of QC and results across the installed base	RC-D	Clinical Affairs	2028-09-30	2028-09-28
CA1	Corrective	Fan control on laser base temperature (FW-311)	RC-1	Firmware	2028-08-30	2028-08-28
CA2	Corrective	Thermal soak in final acceptance test	RC-1	Manufacturing Engineering	2028-09-15	2028-09-14
CA3	Corrective	Laser incoming inspection: firmware and hot test	RC-2	Supplier Quality	2028-08-31	2028-08-30
CA4	Corrective	Quality agreement revision and audit of Vendor L1	RC-2	Supplier Quality	2028-09-30	2028-09-22
CA5	Corrective	Anomaly procedure: environmental causes must be proven	RC-3	Quality	2028-08-31	2028-08-29
CA6	Corrective	New system requirement for thermal margin	RC-1	Systems Engineering	2028-09-15	2028-09-10
CA7	Corrective	Repeat thermal design verification with multiple component lots	RC-1	V&V Engineering	2028-10-15	2028-10-09
CA8	Corrective	Laser temperature visible to users and in QC reports	RC-1, RC-3	Software	2028-09-30	2028-09-24
CA9	Corrective	Firmware version as a controlled attribute of purchased parts	RC-2	Supplier Quality	2028-09-30	2028-09-26
CA10	Corrective	Supplier corrective action request to Vendor L1	RC-2	Supplier Quality	2028-10-31	2028-10-20
CA11	Corrective	Service troubleshooting for QC failures	RC-3	Service	2028-10-15	2028-10-12
PA1	Preventive	Firmware change notification for all critical suppliers	Extent (RC-2)	Supplier Quality	2028-10-31	2028-10-27
PA2	Preventive	Thermal margin review of other components	Extent (RC-1)	Optics R&D	2028-10-15	2028-10-11
PA3	Preventive	Automatic post-market alert on laser temperature	Extent (RC-3)	Service	2028-09-30	2028-09-29
PA4	Preventive	Risk management file and design review checklist update	Extent (RC-1, RC-2)	Quality	2028-09-30	2028-09-27
PA5	Preventive	Training on revised procedures	RC-2, RC-3	Quality	2028-10-31	2028-10-25
PA6	Preventive	DFMEA template: purchased module firmware as a cause	Extent (RC-2)	Systems Engineering	2028-11-15	2028-11-08

ID	Type	Action	Root cause	Owner	Due	Completed
PA7	Preventive	Statistical trending of QC failures per detector	Extent (RC-3)	Quality	2028-11-30	2028-11-21
PA8	Preventive	Site preparation guidance on heat load and airflow	Extent (RC-D)	Service	2028-10-31	2028-10-18
PA9	Preventive	IFU update: QC failure troubleshooting	Extent (RC-3)	Technical Publications	2028-11-15	2028-11-10
PA10	Preventive	Internal audit of supplier controls	Extent (RC-2)	Quality	2028-11-30	2028-11-17
PA11	Preventive	Installed-base firmware inventory	Extent (RC-2)	Firmware	2028-12-15	2028-12-06
PA12	Preventive	Lessons learned to the next program	Extent (RC-1, RC-2, RC-3)	Systems Engineering	2028-12-15	2028-12-01

7.2 Coverage of root causes

Root cause	Actions addressing it
RC-D Direct cause	C1, C2, C3, C4, PA8
RC-1 Design	C1, CA1, CA2, CA6, CA7, CA8, PA2, PA4, PA12
RC-2 Supplier control	CA3, CA4, CA9, CA10, PA1, PA4, PA5, PA6, PA10, PA11, PA12
RC-3 Anomaly procedure and signal detection	CA5, CA8, CA11, PA3, PA5, PA7, PA9, PA12

Every root cause is addressed by at least one corrective action and at least one preventive action that extends the fix beyond the failing part.

7.3 Detailed action definitions

Each action is defined with what will be done, the deliverables, who is responsible, how completion is judged (acceptance criteria, set before the action started) and the evidence that it was met.

C1	Field firmware update 3.0.2 to installed base (Correction)
Root cause addressed	RC-D, RC-1
Description	Install controller firmware 3.0.2 (fan control on laser base temperature, CA1) on all 64 installed instruments with the 405 nm laser, by remote update where the site allows it and by service visit otherwise. Each update ends with a performance check and a record of the firmware version read back from the instrument.
Deliverables	Updated firmware on 64 instruments; field update records FUR-2028-001 to -064; customer notification letter.
Owner / support	Service (support: Firmware, Regulatory)
Due / completed	2028-09-30 / 2028-09-26
Acceptance criteria	100% of the 64 instruments show firmware 3.0.2 in version readback and pass the post-update performance check.
Verification evidence	64 of 64 updated; 64 of 64 passed post-update QC; readback matched release manifest.
Records	CR-SW-0512; FUR-2028-001 to -064
Status	Done

C2	Disposition of held laser lots (Correction)
Root cause addressed	RC-D
Description	Evaluate the 14 held laser modules (lots L1-2801 to L1-2806). Confirm with Vendor L1 that the new TEC firmware is otherwise identical in function, then release the lots for use with firmware 3.0.2 or return them if any module fails the new incoming test (CA3).
Deliverables	NCR disposition; incoming test results for all 14 modules.
Owner / support	Supply Chain (support: Supplier Quality, Optics R&D)
Due / completed	2028-09-15 / 2028-09-12
Acceptance criteria	All released modules pass the 32 °C power stability test in CA3; disposition approved by Quality.
Verification evidence	14 of 14 passed; released under NCR-2028-077 disposition 'use as is with firmware 3.0.2'.
Records	NCR-2028-077
Status	Done

C3	New QC baselines after firmware update (Correction)
Root cause addressed	RC-D
Description	Because the new fan control changes the laser operating temperature, create a new performance check baseline (AD-601) on every updated instrument, so that daily QC compares against the corrected thermal state and not against a baseline taken with the old behavior.
Deliverables	New baseline records for 64 instruments in the Admin Application; service checklist SC-2028-14.
Owner / support	Service
Due / completed	2028-10-07 / 2028-09-30
Acceptance criteria	Baseline created and released for every updated configuration; first daily QC after baseline passes.
Verification evidence	64 baselines released; all first QC runs passed.
Records	SC-2028-14; baseline records in each instrument database
Status	Done

C4	Retrospective review of QC and results across the installed base (Correction)
Root cause addressed	RC-D
Description	Extend the clinical review in Section 2.1 from the 7 complaining sites to all 64 instruments: review violet-detector QC history from installation, identify any runs where QC passed but violet intensity was trending down, and check released IVD results from those days for violet-excited markers against assay controls.
Deliverables	Retrospective review report RR-2028-014.
Owner / support	Clinical Affairs (support: Service, Data Management)
Due / completed	2028-09-30 / 2028-09-28
Acceptance criteria	Every instrument reviewed; any result at risk identified and communicated to the laboratory.
Verification evidence	64 instruments, 9,870 QC runs reviewed; 3 additional sites with warnings found; no released result outside assay control limits.
Records	RR-2028-014
Status	Done

CA1	Fan control on laser base temperature (FW-311) (Corrective)
Root cause addressed	RC-1
Description	Change controller firmware so that fan duty is controlled by the hottest laser base temperature, with a 30 °C setpoint, full duty above 31 °C and an alarm and log entry at 32 °C. Add requirement FW-311 to the software requirements and trace it to the new system requirement in CA6.
Deliverables	Controller firmware 3.0.2; SRS-SW-CSI-010 Rev F (FW-311); updated trace matrix.
Owner / support	Firmware (support: Systems, Optics R&D)
Due / completed	2028-08-30 / 2028-08-28
Acceptance criteria	Laser base temperature ≤ 31 °C at 30 °C room with new-firmware lasers; noise ≤ 62 dBA at full fan duty; no regressions.
Verification evidence	Chamber tests 5 and 6 pass; noise 60 dBA; STS-CSI-010 §B regression pass.
Records	CR-SW-0512; SRS-SW-CSI-010 Rev F
Status	Done

CA2	Thermal soak in final acceptance test (Corrective)
Root cause addressed	RC-1
Description	Add a 6 h soak at 30 °C room with all lasers on to the final acceptance test, followed by a violet and full-detector performance check. Validate the test method and confirm production capacity for the longer test.
Deliverables	FTS-CSI-001 Rev B; WI-CSI-1010 Rev C; test method validation addendum 4.
Owner / support	Manufacturing Engineering
Due / completed	2028-09-15 / 2028-09-14
Acceptance criteria	Method detects the failure reproduced in test 2; production takt time maintained with two additional soak positions.
Verification evidence	TMV addendum 4 pass; 5 production units pass; capacity added.
Records	ECO-0241; FTS-CSI-001 Rev B
Status	Done

CA3	Laser incoming inspection: firmware and hot test (Corrective)
Root cause addressed	RC-2
Description	Revise incoming inspection for all lasers: record firmware version and compare it with the approved list (CA9); test output power stability for 2 h at 32 °C base temperature on 3 modules per lot (lot rejected on any failure).
Deliverables	INS-CSI-001 Rev B; hot-test fixture qualified.
Owner / support	Supplier Quality (support: Optics R&D)
Due / completed	2028-08-31 / 2028-08-30
Acceptance criteria	Seeded unapproved firmware detected; fixture IQ/OQ complete; sampling plan justified (0 failures in 3 per lot).
Verification evidence	3 lots inspected; seeded firmware detected; fixture qualified.
Records	ECO-0244; INS-CSI-001 Rev B
Status	Done

CA4	Quality agreement revision and audit of Vendor L1 (Corrective)
Root cause addressed	RC-2
Description	Revise the quality agreement with Vendor L1 to require 90 days' notice of any change to firmware, software, configuration, materials, process or site, with samples for qualification. Audit Vendor L1's change control process.
Deliverables	QA-L1 Rev C signed; audit report AUD-2028-031.
Owner / support	Supplier Quality
Due / completed	2028-09-30 / 2028-09-22
Acceptance criteria	Agreement signed; audit with no open major findings.
Verification evidence	Signed 2028-09-15; audit 2028-09-20, 2 minor findings closed.
Records	QA-L1 Rev C; SCN-L1-2028-03; AUD-2028-031
Status	Done

CA5	Anomaly procedure: environmental causes must be proven (Corrective)
Root cause addressed	RC-3
Description	Revise the anomaly investigation procedure: an anomaly may be closed as environmental or material-related only when the cause is reproduced, or when instrument causes are excluded with data (for example internal log temperatures, retest with a new lot). Add a closure checklist item and Quality sign-off for such closures.
Deliverables	SOP-QA-017 Rev D; closure checklist form F-QA-017-3.
Owner / support	Quality
Due / completed	2028-08-31 / 2028-08-29
Acceptance criteria	Procedure approved; checklist in the tracking system.
Verification evidence	Approved; tool updated; audited under E6.
Records	SOP-QA-017 Rev D
Status	Done

CA6	New system requirement for thermal margin (Corrective)
Root cause addressed	RC-1
Description	Add a system requirement that every temperature-sensitive component (lasers, PMTs, cameras, electronics) stays within its manufacturer limit with at least 2 °C margin at 32 °C room, and that the margin is verified with components from at least two supplier lots. Allocate to SS-04, SS-05, SS-06 and SS-10.
Deliverables	SRS-CSI-001 Rev E (SYS-327); updated VVP and trace matrix.
Owner / support	Systems Engineering
Due / completed	2028-09-15 / 2028-09-10
Acceptance criteria	Requirement approved at design review; traced to CA1, CA7 and PA2.
Verification evidence	Design review DR-2028-07 approved; TM-CSI-DT-01 Rev D shows full trace.
Records	SRS-CSI-001 Rev E; TM-CSI-DT-01 Rev D
Status	Done

CA7	Repeat thermal design verification with multiple component lots (Corrective)
Root cause addressed	RC-1
Description	Repeat DVP-CSI-301 (operating temperature) and DVP-CSI-403 (excitation) at 30 °C and 32 °C with lasers from the original lot and from the new lots, on 3 instruments with firmware 3.0.2. Revise the V&V plan so that design verification of temperature-sensitive performance uses components from at least two supplier lots.
Deliverables	Verification report addendum PVR-CSI-001 Add. 2; VVP-CSI-001 Rev B.
Owner / support	V&V Engineering
Due / completed	2028-10-15 / 2028-10-09
Acceptance criteria	All requirements in SRS Sections 3.3 and 4 met at 30 °C; SYS-327 met at 32 °C.
Verification evidence	All pass; lowest margin 2.6 °C (561 nm laser).
Records	PVR-CSI-001 Add. 2; VVP-CSI-001 Rev B
Status	Done

CA8	Laser temperature visible to users and in QC reports (Corrective)
Root cause addressed	RC-1, RC-3
Description	Show laser base temperatures in the Research Application status panel and in the performance check report, and add a QC warning when any laser is above 31 °C during the check. This gives users and service a direct clue instead of suspecting the bead lot first.
Deliverables	Research Application 3.0.2 (RA-107); firmware log entries (FW-312).
Owner / support	Software (support: Human Factors)
Due / completed	2028-09-30 / 2028-09-24
Acceptance criteria	Values displayed and reported; warning triggers at 31 °C in chamber test; formative check with 5 users shows correct interpretation.
Verification evidence	STS \$A regression pass; 5 of 5 users interpreted warning correctly.
Records	CR-SW-0514; SRS-SW-CSI-010 Rev F
Status	Done

CA9	Firmware version as a controlled attribute of purchased parts (Corrective)
Root cause addressed	RC-2
Description	Record approved firmware versions for every purchased part that contains firmware as a controlled attribute in the bill of materials and approved vendor list, so that a new version requires an engineering change and qualification before use.
Deliverables	BOM-CSI-0001 Rev D; AVL-CSI-001 Rev C; PLM attribute 'approved firmware'.
Owner / support	Supplier Quality (support: Configuration Management)
Due / completed	2028-09-30 / 2028-09-26
Acceptance criteria	All 11 part numbers with firmware have approved versions recorded; PLM blocks release of parts without the attribute.
Verification evidence	11 of 11 recorded; PLM rule tested.
Records	ECO-0246; AVL-CSI-001 Rev C
Status	Done

CA10	Supplier corrective action request to Vendor L1 (Corrective)
Root cause addressed	RC-2
Description	Issue a SCAR requiring Vendor L1 to investigate why the TEC firmware change was not notified, correct its change classification rules and confirm that no other undisclosed changes affect parts supplied to CSI.
Deliverables	SCAR-2028-009 response with root cause and corrective action.
Owner / support	Supplier Quality
Due / completed	2028-10-31 / 2028-10-20
Acceptance criteria	Supplier root cause accepted; change classification corrected; declaration of no other undisclosed changes.
Verification evidence	Response accepted 2028-10-20; verified at audit follow-up.
Records	SCAR-2028-009
Status	Done

CA11	Service troubleshooting for QC failures (Corrective)
Root cause addressed	RC-3
Description	Revise the service manual troubleshooting tree for QC failures: check laser temperatures and internal log before replacing bead lots or realigning; add a data collection step (log export) for every QC-failure service call.
Deliverables	SM-CSI-001 Rev C; service training module ST-2028-06.
Owner / support	Service
Due / completed	2028-10-15 / 2028-10-12
Acceptance criteria	Manual released; all field engineers trained.
Verification evidence	Released; 22 of 22 field engineers trained.
Records	SM-CSI-001 Rev C
Status	Done

PA1	Firmware change notification for all critical suppliers (Preventive)
Root cause addressed	Extent (RC-2)
Description	Extend firmware, software and configuration change notification to every critical supplier whose parts contain firmware (lasers, cameras, HV supply, PMT modules, water bath, fluidics controller board). Revise the supplier agreement template and the supplier control procedure.
Deliverables	SOP-SQ-004 Rev E; supplier agreement template Rev D; 7 revised agreements.
Owner / support	Supplier Quality
Due / completed	2028-10-31 / 2028-10-27
Acceptance criteria	All 7 agreements signed.
Verification evidence	7 of 7 signed; 3 notifications already received (E5).
Records	SOP-SQ-004 Rev E
Status	Done

PA2	Thermal margin review of other components (Preventive)
Root cause addressed	Extent (RC-1)
Description	Analyze and test thermal margin of the 488, 561 and 633 nm lasers, PMTs, cameras and HV supply at 32 °C room with worst-case tolerances and airflow, and confirm fan control 3.0.2 keeps each within limit.
Deliverables	Technical report TR-CSI-2028-09.
Owner / support	Optics R&D (support: Electrical)
Due / completed	2028-10-15 / 2028-10-11
Acceptance criteria	Margin ≥ 2 °C for every component (SYS-327).
Verification evidence	Lowest margin 2.6 °C.
Records	TR-CSI-2028-09
Status	Done

PA3	Automatic post-market alert on laser temperature (Preventive)
Root cause addressed	Extent (RC-3)
Description	Add laser base temperature and fan duty to the post-market log trend with an automatic alert to Service when any instrument exceeds 32 °C.
Deliverables	PMS-CSI-001 Rev B; alert rule in service analytics.
Owner / support	Service
Due / completed	2028-09-30 / 2028-09-29
Acceptance criteria	Alert fires on seeded data; reviewed monthly.
Verification evidence	Seeded test pass; first monthly review done.
Records	PMS-CSI-001 Rev B
Status	Done

PA4	Risk management file and design review checklist update (Preventive)
Root cause addressed	Extent (RC-1, RC-2)
Description	Add 'supplier firmware change' and 'thermal margin loss' as causes under HA-08 and review their controls; add thermal margin and supplier firmware questions to the design review checklist.
Deliverables	RMF-CSI-001 Rev D; design review checklist Rev C.
Owner / support	Quality
Due / completed	2028-09-30 / 2028-09-27
Acceptance criteria	Updates approved; residual risk re-evaluated.
Verification evidence	Approved; HA-08 residual unchanged (S4, P2) with added controls.
Records	RMF-CSI-001 Rev D
Status	Done

PA5	Training on revised procedures (Preventive)
Root cause addressed	RC-2, RC-3
Description	Train R&D, Quality, Supplier Quality and Service staff on SOP-QA-017 Rev D, SOP-SQ-004 Rev E and the revised inspection.
Deliverables	Training records TR-2028-114.
Owner / support	Quality
Due / completed	2028-10-31 / 2028-10-25
Acceptance criteria	100% of affected staff trained; quiz score $\geq$ 80%.
Verification evidence	38 of 38 trained; lowest score 85%.
Records	TR-2028-114
Status	Done

PA6	DFMEA template: purchased module firmware as a cause (Preventive)
Root cause addressed	Extent (RC-2)
Description	Add 'change or defect in purchased module firmware' as a standard cause category in the DFMEA template, and apply it to every CSI-100 subsystem with purchased firmware.
Deliverables	DFMEA template Rev F; DFMEA-CSI-001 Rev D.
Owner / support	Systems Engineering
Due / completed	2028-11-15 / 2028-11-08
Acceptance criteria	Template released; all 6 affected subsystems re-analyzed.
Verification evidence	6 of 6 done; no new unacceptable risks.
Records	DFMEA-CSI-001 Rev D
Status	Done

PA7	Statistical trending of QC failures per detector (Preventive)
Root cause addressed	Extent (RC-3)
Description	Replace the count-based CAPA trigger for QC failures with statistical trending (CUSUM per detector array per month across the installed base), so that a shift is detected earlier than 3 complaints.
Deliverables	PMS-CSI-001 Rev B trending method; validated script.
Owner / support	Quality (support: Service analytics)
Due / completed	2028-11-30 / 2028-11-21
Acceptance criteria	Retrospective run on 2028 data signals within 4 weeks of first failures (vs 10 weeks actual).
Verification evidence	Signal at week 3 on retrospective data; script validated.
Records	PMS-CSI-001 Rev B; software tool validation STV-2028-05
Status	Done

PA8	Site preparation guidance on heat load and airflow (Preventive)
Root cause addressed	Extent (RC-D)
Description	Update the site preparation guide with room heat load per configuration, minimum clearances for airflow and a recommendation to confirm HVAC capacity where several instruments share a room.
Deliverables	SPG-CSI-001 Rev B.
Owner / support	Service (support: Mechanical)
Due / completed	2028-10-31 / 2028-10-18
Acceptance criteria	Guide released; used for all new installations.
Verification evidence	Released; applied to 9 installations in the period.
Records	SPG-CSI-001 Rev B
Status	Done

PA9	IFU update: QC failure troubleshooting (Preventive)
Root cause addressed	Extent (RC-3)
Description	Add a QC troubleshooting section to the Operator's Guide that shows the laser temperature indicator (CA8) and tells users when to contact service instead of changing bead lots.
Deliverables	OG-CSI-001 Rev D.
Owner / support	Technical Publications (support: Regulatory)
Due / completed	2028-11-15 / 2028-11-10
Acceptance criteria	Labeling review complete; translations released.
Verification evidence	Approved; 6 languages released.
Records	OG-CSI-001 Rev D; DCR-IFU-031
Status	Done

PA10	Internal audit of supplier controls (Preventive)
Root cause addressed	Extent (RC-2)
Description	Audit the supplier control process (ISO 13485 clause 7.4) across all product lines, with focus on change notification and incoming inspection of parts with firmware.
Deliverables	Internal audit report IA-2028-12.
Owner / support	Quality (internal audit)
Due / completed	2028-11-30 / 2028-11-17
Acceptance criteria	Audit complete; findings in CAPA or correction.
Verification evidence	2 minor findings, corrected under the audit.
Records	IA-2028-12
Status	Done

PA11	Installed-base firmware inventory (Preventive)
Root cause addressed	Extent (RC-2)
Description	Report firmware versions of all purchased modules from each instrument to the service portal at every service connection, so that the installed base can be searched when a supplier notifies a change.
Deliverables	Firmware 3.0.3 (FW-313); service portal report.
Owner / support	Firmware (support: Service)
Due / completed	2028-12-15 / 2028-12-06
Acceptance criteria	Inventory available for $\geq 90\%$ of connected instruments.
Verification evidence	61 of 64 instruments reporting (3 sites without connection).
Records	CR-SW-0521
Status	Done

PA12	Lessons learned to the next program (Preventive)
Root cause addressed	Extent (RC-1, RC-2, RC-3)
Description	Record the lessons in the engineering lessons-learned register and add thermal margin, multi-lot verification and supplier firmware control requirements to the requirements template used by the next instrument program (CSI-200).
Deliverables	Lessons learned LL-2028-07; requirements template Rev C.
Owner / support	Systems Engineering
Due / completed	2028-12-15 / 2028-12-01
Acceptance criteria	Items present in CSI-200 SRS draft.
Verification evidence	Confirmed in CSI-200 SRS draft Rev A.
Records	LL-2028-07
Status	Done

8. Verification of Actions

Each action was verified against its acceptance criteria before it was counted as complete. Design changes (CA1, CA6, CA7, CA8, PA11) were verified through design control, including regression testing and a risk review; production and supplier changes (CA2, CA3, CA9) were validated before release to production.

Action	Acceptance criteria	Result
C1 Field firmware update 3.0.2 to installed base	100% of the 64 instruments show firmware 3.0.2 in version readback and pass the post-update performance check.	Met: 64 of 64 updated; 64 of 64 passed post-update QC; readback matched release manifest.
C2 Disposition of held laser lots	All released modules pass the 32 °C power stability test in CA3; disposition approved by Quality.	Met: 14 of 14 passed; released under NCR-2028-077 disposition 'use as is with firmware 3.0.2'.
C3 New QC baselines after firmware update	Baseline created and released for every updated configuration; first daily QC after baseline passes.	Met: 64 baselines released; all first QC runs passed.
C4 Retrospective review of QC and results across the installed base	Every instrument reviewed; any result at risk identified and communicated to the laboratory.	Met: 64 instruments, 9,870 QC runs reviewed; 3 additional sites with warnings found; no released result outside assay control limits.
CA1 Fan control on laser base temperature (FW-311)	Laser base temperature ≤ 31 °C at 30 °C room with new-firmware lasers; noise ≤ 62 dBA at full fan duty; no regressions.	Met: Chamber tests 5 and 6 pass; noise 60 dBA; STS-CSI-010 §B regression pass.
CA2 Thermal soak in final acceptance test	Method detects the failure reproduced in test 2; production takt time maintained with two additional soak positions.	Met: TMV addendum 4 pass; 5 production units pass; capacity added.
CA3 Laser incoming inspection: firmware and hot test	Seeded unapproved firmware detected; fixture IQ/OQ complete; sampling plan justified (0 failures in 3 per lot).	Met: 3 lots inspected; seeded firmware detected; fixture qualified.
CA4 Quality agreement revision and audit of Vendor L1	Agreement signed; audit with no open major findings.	Met: Signed 2028-09-15; audit 2028-09-20, 2 minor findings closed.
CA5 Anomaly procedure: environmental causes must be proven	Procedure approved; checklist in the tracking system.	Met: Approved; tool updated; audited under E6.
CA6 New system requirement for thermal margin	Requirement approved at design review; traced to CA1, CA7 and PA2.	Met: Design review DR-2028-07 approved; TM-CSI-DT-01 Rev D shows full trace.
CA7 Repeat thermal design verification with multiple component lots	All requirements in SRS Sections 3.3 and 4 met at 30 °C; SYS-327 met at 32 °C.	Met: All pass; lowest margin 2.6 °C (561 nm laser).
CA8 Laser temperature visible to users and in QC reports	Values displayed and reported; warning triggers at 31 °C in chamber test; formative check with 5 users shows correct interpretation.	Met: STS §A regression pass; 5 of 5 users interpreted warning correctly.
CA9 Firmware version as a controlled attribute of purchased parts	All 11 part numbers with firmware have approved versions recorded; PLM blocks release of parts without the attribute.	Met: 11 of 11 recorded; PLM rule tested.
CA10 Supplier corrective action request to Vendor L1	Supplier root cause accepted; change classification corrected; declaration of no other undisclosed changes.	Met: Response accepted 2028-10-20; verified at audit follow-up.
CA11 Service troubleshooting for QC failures	Manual released; all field engineers trained.	Met: Released; 22 of 22 field engineers trained.
PA1 Firmware change notification for all critical suppliers	All 7 agreements signed.	Met: 7 of 7 signed; 3 notifications already received (E5).

Action	Acceptance criteria	Result
PA2 Thermal margin review of other components	Margin ≥ 2 °C for every component (SYS-327).	Met: Lowest margin 2.6 °C.
PA3 Automatic post-market alert on laser temperature	Alert fires on seeded data; reviewed monthly.	Met: Seeded test pass; first monthly review done.
PA4 Risk management file and design review checklist update	Updates approved; residual risk re-evaluated.	Met: Approved; HA-08 residual unchanged (S4, P2) with added controls.
PA5 Training on revised procedures	100% of affected staff trained; quiz score $\geq 80\%$.	Met: 38 of 38 trained; lowest score 85%.
PA6 DFMEA template: purchased module firmware as a cause	Template released; all 6 affected subsystems re-analyzed.	Met: 6 of 6 done; no new unacceptable risks.
PA7 Statistical trending of QC failures per detector	Retrospective run on 2028 data signals within 4 weeks of first failures (vs 10 weeks actual).	Met: Signal at week 3 on retrospective data; script validated.
PA8 Site preparation guidance on heat load and airflow	Guide released; used for all new installations.	Met: Released; applied to 9 installations in the period.
PA9 IFU update: QC failure troubleshooting	Labeling review complete; translations released.	Met: Approved; 6 languages released.
PA10 Internal audit of supplier controls	Audit complete; findings in CAPA or correction.	Met: 2 minor findings, corrected under the audit.
PA11 Installed-base firmware inventory	Inventory available for $\geq 90\%$ of connected instruments.	Met: 61 of 64 instruments reporting (3 sites without connection).
PA12 Lessons learned to the next program	Items present in CSI-200 SRS draft.	Met: Confirmed in CSI-200 SRS draft Rev A.

All 27 actions met their acceptance criteria. The risk review of CA1 and CA8 found no new hazards; increased fan duty stays below the noise limit (60 dBA against 62 dBA).

9. Effectiveness Check

Effectiveness criteria were defined and approved on 2028-08-05, before implementation, so that success could not be redefined after the data were seen. The check period was 90 days after the last field update (2028-09-26 to 2028-12-25; data frozen 2028-12-12 when all criteria were already met with the required sample).

ID	Criterion	Target	Result	Status
E1	Violet QC failures or warnings attributable to laser temperature on updated instruments	0 in 90 days across all 64 instruments ($\geq 5,000$ QC runs)	0 in 5,412 QC runs	Met
E2	Complaint rate for violet QC failure	≤ 0.1 per 100 instrument-months (pre-CAPA 5.1)	0.0 (0 complaints, 176 instrument-months)	Met
E3	Laser base temperature from internal logs at room ≥ 27 °C	99th percentile ≤ 31 °C	99th percentile 30.4 °C (1,930 h at ≥ 27 °C)	Met
E4	Incoming inspection detects unapproved firmware	100% of lots checked; 0 escapes	6 lots checked; 1 lot with new firmware held and qualified before release	Met
E5	Supplier notifications received for firmware changes	Notifications received under revised agreements	3 notifications received from 2 suppliers; all assessed	Met
E6	Anomaly closures with environmental cause include reproduction or exclusion data	100% of closures audited in the period	12 of 12	Met

Violet QC failures or warnings per 10 affected-lot instruments per month, 2028

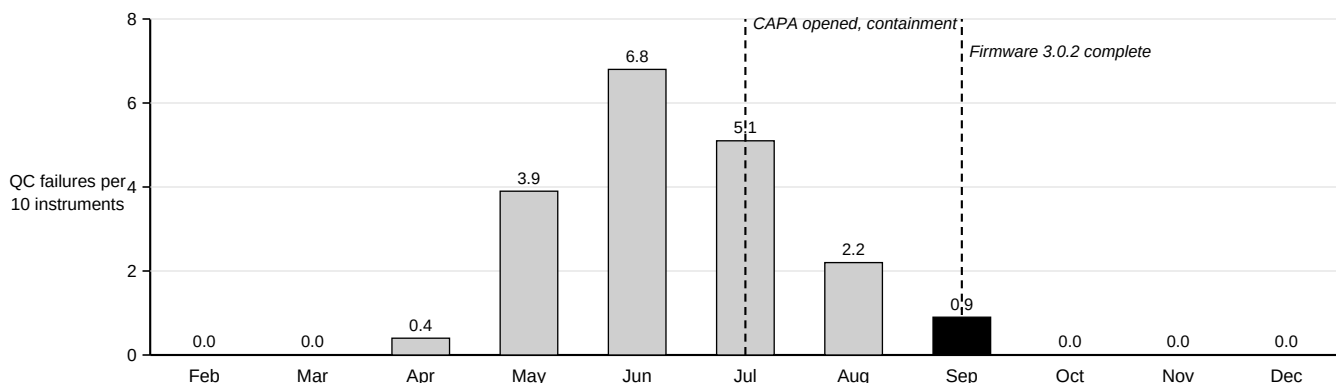


Figure 3. Violet QC failures before and after the CAPA. Grey: before full correction; black: effectiveness check period.

Conclusion: all six effectiveness criteria were met. The problem has not recurred, the supplier and anomaly process changes are working, and no new problems were introduced.

10. Closure

Closure check	Evidence	Complete
Root cause confirmed by test	Section 5	Yes
All actions implemented and verified	Sections 7 and 8	Yes
Design changes processed through design control; DHF updated	CR-SW-0512, -0514, -0521; SRS-CSI-001 Rev E; SRS-SW-CSI-010 Rev F; PVR-CSI-001 Add. 2; TM-CSI-DT-01 Rev D	Yes
DMR and production updated (design transfer)	ECO-0241, -0244, -0246; FTS-CSI-001 Rev B; INS-CSI-001 Rev B; BOM-CSI-0001 Rev D; AVL-CSI-001 Rev C	Yes
Supplier actions complete	QA-L1 Rev C; SCAR-2028-009 accepted; 7 agreements under PA1; audit IA-2028-12	Yes
Labeling, service and site documents updated	OG-CSI-001 Rev D; SM-CSI-001 Rev C; SPG-CSI-001 Rev B	Yes
Extent of issue addressed (other components, suppliers, products)	PA1, PA2, PA6, PA10, PA12	Yes
Risk management file updated	RMF-CSI-001 Rev D	Yes
Reportability decisions documented and re-confirmed	Section 2.3	Yes
Effectiveness criteria met	Section 9	Yes
Customers informed of correction	TB-CSI-2028-02; field update records FUR-2028-001 to -064; retrospective review RR-2028-014	Yes
Input to management review	Management review 2029-Q1 agenda item 4	Yes

Closure statement: CAPA-2028-014 is closed as effective. The violet detector QC failures were caused by a supplier firmware change combined with fan control that did not follow laser temperature. The design, supplier controls, inspection and anomaly procedures have been corrected, and the effectiveness check showed no recurrence across the installed base.

Role	Name	Signature / Date
CAPA owner	A. Lodhi, Systems Engineering	2028-12-15
Quality Engineering	Quality Manager	2028-12-15
Regulatory Affairs	Regulatory Manager	2028-12-15
Management representative	VP Quality and Regulatory	2028-12-15

11. Records and References

Record	Description
CMP-2028-0031 to -0066 (7 records)	Customer complaints
CAPA-2028-014-T1	Root cause confirmation test protocol and report
TB-CSI-2028-02	Customer technical bulletin
NCR-2028-077	Nonconformance report for held laser lots
CR-SW-0512; SRS-SW-CSI-010 Rev F	Firmware change (FW-311 fan control) and requirement update
ECO-0241; FTS-CSI-001 Rev B	Thermal soak added to final acceptance test
ECO-0244; INS-CSI-001 Rev B	Laser incoming inspection revision
QA-L1 Rev C; SCN-L1-2028-03; SOP-SQ-004 Rev E	Supplier quality agreement, change notification, supplier control procedure
SOP-QA-017 Rev D	Anomaly investigation procedure
TR-CSI-2028-09	Thermal margin review for other components
RMF-CSI-001 Rev D; PMS-CSI-001 Rev B	Risk management file; post-market surveillance plan
ADL-CSI-001 (ANM-0062)	Earlier validation anomaly reviewed in this CAPA
FUR-2028-001 to -064; SC-2028-14; RR-2028-014	Field update records; baseline checklist; retrospective review
SRS-CSI-001 Rev E; PVR-CSI-001 Add. 2; VVP-CSI-001 Rev B; TM-CSI-DT-01 Rev D	New thermal requirement, repeated verification, V&V plan and trace updates
CR-SW-0514, CR-SW-0521; ECO-0246; AVL-CSI-001 Rev C; BOM-CSI-0001 Rev D	Laser temperature display, firmware inventory, controlled firmware attribute
SCAR-2028-009; AUD-2028-031; IA-2028-12	Supplier corrective action, supplier audit, internal audit
SM-CSI-001 Rev C; SPG-CSI-001 Rev B; OG-CSI-001 Rev D	Service manual, site preparation guide, Operator Guide
DFMEA-CSI-001 Rev D; PMS-CSI-001 Rev B; STV-2028-05; LL-2028-07	DFMEA, post-market trending method and tool validation, lessons learned
ISO 13485:2016 8.5.2, 8.5.3; 21 CFR 820; 21 CFR 803, 806; EU IVDR Art. 82 to 83	Requirements for corrective and preventive action and reporting